

1 Reminder on the Lagrangian Multipliers

$$\begin{array}{ll} \text{minimize} & f_0(\mathbf{x}) \\ \text{subject to} & f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & h_i(\mathbf{x}) = 0, \quad i = 1, \dots, p \end{array}$$

We take the constraints and add them as regularizer terms to the objective; making this an unconstrained problem. This forms the Lagrangian:

$$\mathfrak{L}(x, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i h_i(\mathbf{x}),$$

Note that Lagrangian is a function of both primal variable (x) and duals (λ , ν). Next, we minimize the Lagrangian with respect to the primal variable to get the Lagrange dual function (or just dual function):

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{x \in \mathcal{D}} \mathfrak{L}(x, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{x \in \mathcal{D}} \left(f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i h_i(\mathbf{x}) \right)$$

Notice that since we optimized over the primal $\mathbf{x}$, the dual function has no longer any dependency on it.

Notice that we have for any point $\tilde{\mathbf{x}}$ that satisfies the constraints (feasible) we have that $L(\tilde{\mathbf{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f_0(\tilde{\mathbf{x}})$.

But since we minimized the Lagrangian, that function should always be a lower bound on the Lagrangian as well; so:

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{x \in \mathcal{D}} L(x, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\tilde{\mathbf{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f_0(\tilde{\mathbf{x}}).$$

This includes $\mathbf{x}^* = \text{argmin}(f_0)$ which gives $p^* = f_0(\mathbf{x}^*)$ (the optimal loss)

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*$$

So how do we calculate the optimal loss using the dual? We maximize it!

2 SVM Primal & Dual Problems

Previously we have seen the SVM primal problem as:

$$\begin{array}{ll} \min_{\mathbf{w}, b, \boldsymbol{\xi}} & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n \\ \text{subject to} & y_n (\langle \mathbf{w}, \mathbf{x}_n \rangle + b) \geq 1 - \xi_n \end{array} \quad (1)$$

$$\xi_n \geq 0 \quad (2)$$

Let us write the Lagrangian:

$$\begin{aligned} \mathfrak{L}(\mathbf{w}, b, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\gamma}) = & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n \\ & - \underbrace{\sum_{n=1}^N \alpha_n (y_n (\langle \mathbf{w}, \mathbf{x}_n \rangle + b) - 1 + \xi_n)}_{\text{constraint (1)}} - \underbrace{\sum_{n=1}^N \gamma_n \xi_n}_{\text{constraint (2)}}. \end{aligned}$$

Optimizing with respect to the primal variables: $\mathbf{w}, b, \xi$:

$$\frac{\partial \mathfrak{L}}{\partial \mathbf{w}} = \mathbf{w} - \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n, \quad (3)$$

$$\frac{\partial \mathfrak{L}}{\partial b} = - \sum_{n=1}^N \alpha_n y_n, \quad (4)$$

$$\frac{\partial \mathfrak{L}}{\partial \xi_n} = C - \alpha_n - \gamma_n. \quad \forall n \in [N] \quad (5)$$

Setting Eq.(3) to zero, we have:

$$\mathbf{w} = \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n$$

Representer Theorem: Optimal weight vector in the primal is a linear combination of the examples $\mathbf{x}_n$.

- The solution to the optimization problem lies in the span of training data
- The provenance of the name: Support vector machine
- The examples $\mathbf{x}_n$, for which the corresponding parameters $\alpha_n = 0$, do not contribute to the solution $\mathbf{w}$ at all. The other examples, where $\alpha_n > 0$, are called support vectors since they “support” the hyperplane.

Working out the dual objective problem

$$\begin{aligned} \tilde{\mathfrak{L}}(\boldsymbol{\alpha}, \boldsymbol{\gamma}) = & \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y_i y_j \alpha_i \alpha_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle - \sum_{i=1}^N y_i \alpha_i \left\langle \sum_{j=1}^N y_j \alpha_j \mathbf{x}_j, \mathbf{x}_i \right\rangle \\ & + C \sum_{i=1}^N \xi_i - b \sum_{i=1}^N y_i \alpha_i + \sum_{i=1}^N \alpha_i - \sum_{i=1}^N \alpha_i \xi_i - \sum_{i=1}^N \gamma_i \xi_i. \end{aligned} \quad (6)$$

- Not a function of primal parameters $\mathbf{w}, b$ anymore.
- Figuring out the constraints:

- Setting Eq. (4) to zero gives us the equality constraint $\sum_{n=1}^N y_n \alpha_n = 0$
- Setting Eq. (5) to zero gives us $C - \alpha_n = \gamma_n$ but γ_n is a Lagrange variable so $\gamma_n \geq 0$. Therefore,

$$0 \leq \alpha_i \leq C \text{ for all } i = 1, \dots, N$$

Back to Eq. (6):

- The first two terms are the same (we can pull the summation out of the 2nd inner product)
- 4'th term is zero by the equality constraint $\sum_{n=1}^N y_n \alpha_n = 0$
- We can factor out ξ out of all the remaining terms to get

$$\tilde{\mathcal{L}}(\xi, \alpha, \gamma) = -\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y_i y_j \alpha_i \alpha_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle + \sum_{i=1}^N \alpha_i + \sum_{i=1}^N (C - \alpha_i - \gamma_i) \xi_i.$$

- The third term is also zero (via setting Eq. (5) to zero).

Dual Problem: We have our dual problem as follows:

$$\begin{aligned} \max_{\alpha} \quad & -\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y_i y_j \alpha_i \alpha_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle + \sum_{i=1}^N \alpha_i \\ \text{subject to} \quad & \sum_{i=1}^N y_i \alpha_i = 0 \\ & 0 \leq \alpha_i \leq C \quad \text{for all } i = 1, \dots, N. \end{aligned}$$

- Note that the dual problem is always an opposite optimization problem (if we minimize the primal, we maximize the dual). Why? (The reminder on Lagrangian explains why)
- The inequality constraint is particularly easy to implement in software; hence why we prefer the dual problem for SVM.

Recovering the primal optimal values

- For weights: use the representer theorem
- For bias b :
 - Should not be necessary if data is centered (bias = 0)
 - If there is a point that lies exactly on the boundary, we can use
$$b^* = y_n - \langle \mathbf{w}^*, \mathbf{x}_n \rangle$$
 - Otherwise, we should compute $b^* = y_n - \langle \mathbf{w}^*, \mathbf{x}_n \rangle$ for all support vectors and take the median value of this absolute value difference to be the optimal b^* .